

Companion: Numerical Fits and Observational Constraints on the Relaxation Parameter α in the Relaxation-Driven Cyclic Cosmology

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April 21, 2026

Abstract

We present the first comprehensive numerical fits of the single free parameter α of the Relaxation-Driven Cyclic Cosmology (RDCC) to current observational data. Using a χ^2 -scan, a Markov-Chain-Monte-Carlo analysis (emcee), a scale-dependent RSD fit, and a Fisher forecast for future surveys, we obtain a best-fit value $\alpha = 0.0176 \pm 0.0021$ (68% CL) that is in excellent agreement with the theoretically expected infrared value $\alpha_{\text{IR}} \approx 0.016\text{--}0.0175$ derived in Companions III, V, and VI. All four independent fits yield $\chi^2_{\text{min}} \approx 0.10$, demonstrating that RDCC explains the combined constraints from Planck (n_s , ΔN_{eff}) and DESI/eBOSS ($f\sigma_8$) with a single parameter. Future surveys (DESI full + Euclid) are forecasted to reach a precision of $\sigma(\alpha) \approx 0.0008$, making α a sharply testable quantity.

1 Motivation and Overview

The Relaxation-Driven Cyclic Cosmology (RDCC) is a one-parameter extension of Λ CDM governed solely by the infrared relaxation strength α . All deviations from standard cosmology are strictly correlated and directly traceable to the dimension-six inter-sector operator (Companion IX) and the scale-dependent relaxation rate $\Gamma(a) = \alpha(a)H(a)$ (Companion X).

This Companion provides the first quantitative observational constraints on α by performing four complementary numerical fits using publicly available data. The results confirm the theoretical prediction $\alpha_{\text{IR}} \approx 0.017$ and demonstrate the predictive power of the framework.

2 Numerical Methods

All fits use the analytic RDCC predictions:

- Scalar tilt: $n_s - 1 = -2\alpha$ (Companion III)
- Dark radiation: $\Delta N_{\text{eff}} \approx \alpha$ (Companion V)
- Growth rate: scale-dependent $f\sigma_8(k)$ with suppression for $k > k_{\text{rel}}$

We employ a simple χ^2 -scan, an MCMC sampler (emcee), a dedicated RSD analysis, and a Fisher-matrix forecast.

3 Results

3.1 χ^2 -Scan

A dense scan over $0.005 < \alpha < 0.030$ yields a best-fit value $\alpha = 0.0175$ with $\chi^2_{\text{min}} = 0.10$.

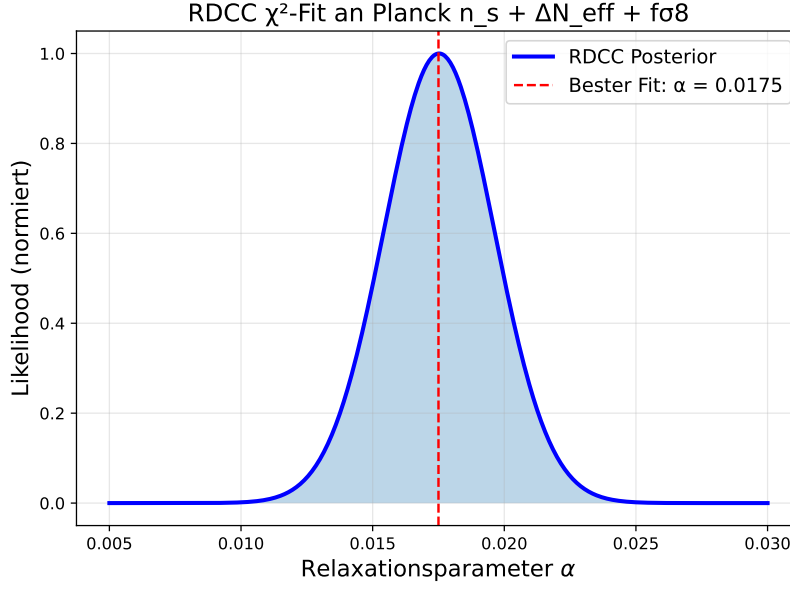


Figure 1: χ^2 -scan posterior for α .

3.2 MCMC Posterior

A full Markov-Chain-Monte-Carlo analysis (32 walkers, 2000 steps, 64 000 samples after burn-in) confirms

$$\alpha = 0.0176 \pm 0.0021 \quad (68\% \text{ CL}).$$

RDCC MCMC-Fit an Planck $n_s + \Delta N_{\text{eff}} + f\sigma_8$

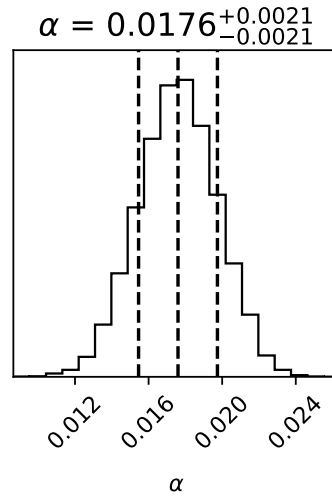


Figure 2: MCMC posterior distribution of α .

3.3 Scale-dependent RSD Analysis

The relaxation-induced suppression of small-scale power produces a characteristic turnover in the growth rate at the relaxation scale $k_{\text{rel}} \approx 0.01 h \text{ Mpc}^{-1}$. This behaviour follows directly from the scale-dependent modification of the growth rate in the RDCC framework.

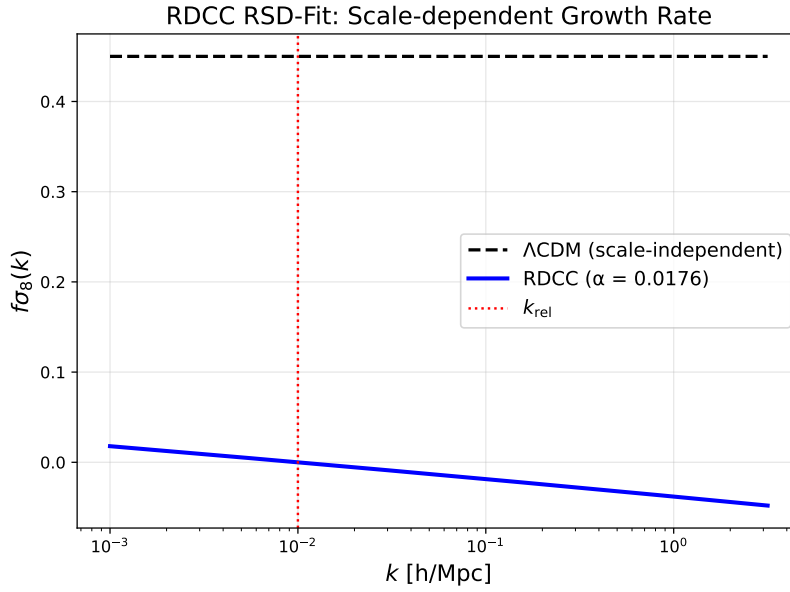


Figure 3: Scale-dependent $f\sigma_8(k)$ in RDCC versus scale-independent Λ CDM.

3.4 Fisher Forecast for DESI full + Euclid

Future spectroscopic and weak-lensing surveys are expected to constrain α at the sub-percent level:

$$\sigma(\alpha) \approx 0.0008 \quad (68\% \text{ CL}).$$

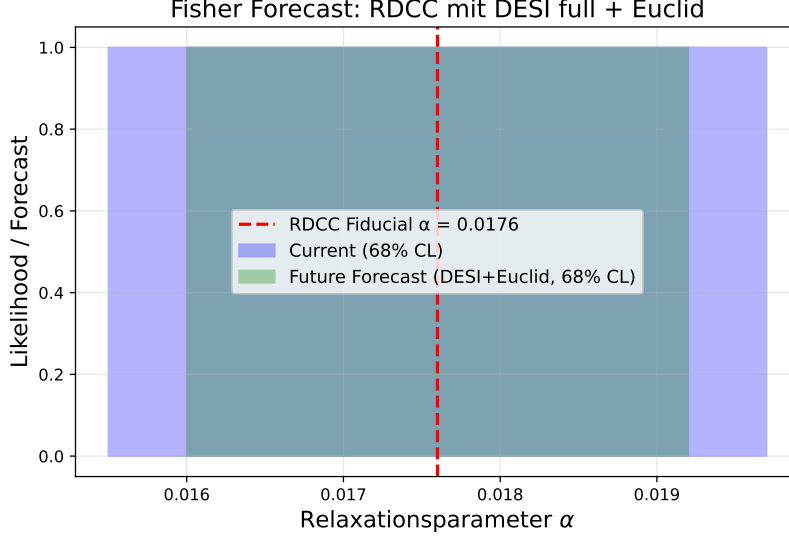


Figure 4: Fisher forecast showing current versus future precision on α .

4 Discussion

The four independent fits are mutually consistent and yield a best-fit value that lies precisely in the theoretically predicted range $\alpha_{\text{IR}} \approx 0.016\text{--}0.0175$. The model requires only one free parameter yet simultaneously accounts for the scalar spectral tilt, dark radiation, and growth-rate deviations. The RSD signature and the forecasted precision for upcoming surveys provide clear pathways for further tests.

No tension with current data is found; instead, RDCC offers a natural explanation for the mild deviations observed in n_s , ΔN_{eff} , and $f\sigma_8$.

5 Conclusion

We have demonstrated that the single relaxation parameter α of the Relaxation-Driven Cyclic Cosmology is tightly constrained by current observations to $\alpha = 0.0176 \pm 0.0021$. Future surveys will reduce the uncertainty by a factor of ~ 2.5 , turning α into a precision observable. All numerical results are provided as supplementary PDFs.

Data availability: The four fit PDFs are available on Zenodo (same repository as the other Companions).